# Practice workbook · Recall the elements and building blocks of food molecules

Chemical make-up of carbohydrates, proteins and lipids: their elements and the small units that join to build them.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-NUTRIENT-COMPOSITION-RECALL

# 1. Question 3 · part a.i · 1 mark

June 2021 Paper 1

Worked-example retry · Attempt part a.i only.

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**3** Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

**(1)**

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

**(3)**

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## 1. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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## 2. Question 3 · part a · 3 marks

June 2019 Paper 2

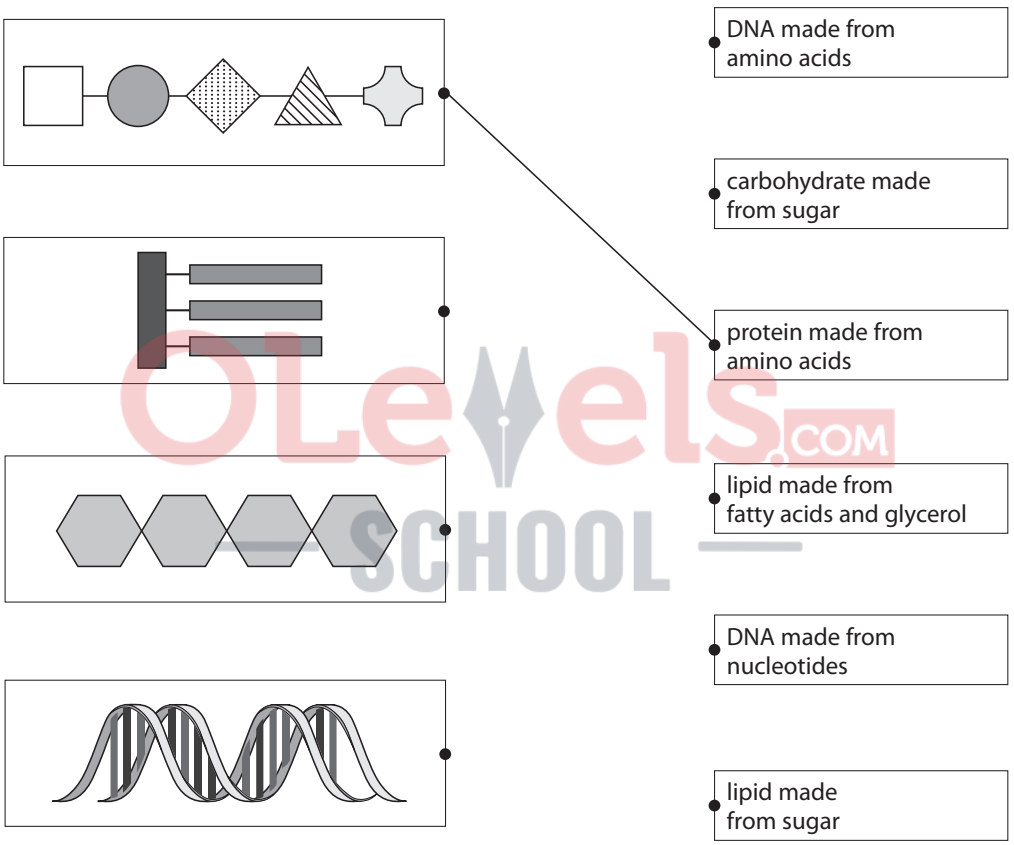
Further practice · Attempt part a only.

**3** (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

**(3)**




(b) Which body organ produces enzymes that break down protein?

**(1)**

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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## 2. Question 3 · part a · 3 marks · Working and answer

Working and answer

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# 3. Question 2 · part a.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.ii only.

**2** A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.

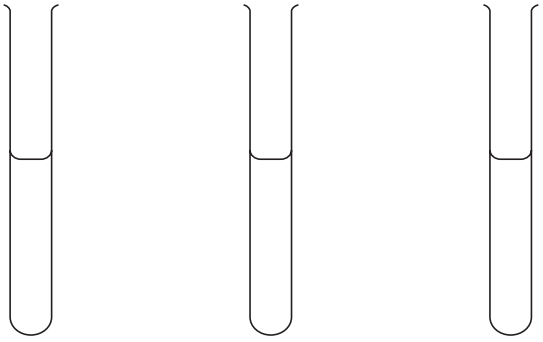



Table 1 lists the contents of each test tube.

| Test tube A | Test tube B | Test tube C |
|---|---|---|
| 5 $cm^3$ of milk | 5 $cm^3$ of milk | 5 $cm^3$ of milk |
| 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate |
| 6 drops of phenolphthalein | 6 drops of phenolphthalein | 6 drops of phenolphthalein |
| 1 $cm^3$ enzyme | 1 $cm^3$ enzyme | 1 $cm^3$ boiled enzyme |
| distilled water | bile salts | bile salts |

**Table 1**

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## 3. Question 2 · part a.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.ii only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

| | Colour of contents | | |
|---|---|---|---|
| | **Tube A** | **Tube B** | **Tube C** |
| at start | pink | pink | pink |
| after 5 minutes | pink | colourless | pink |
| after 10 minutes | pink | colourless | pink |
| after 15 minutes | colourless | colourless | pink |

**Table 2**

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation? **(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids? **(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced? **(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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## 3. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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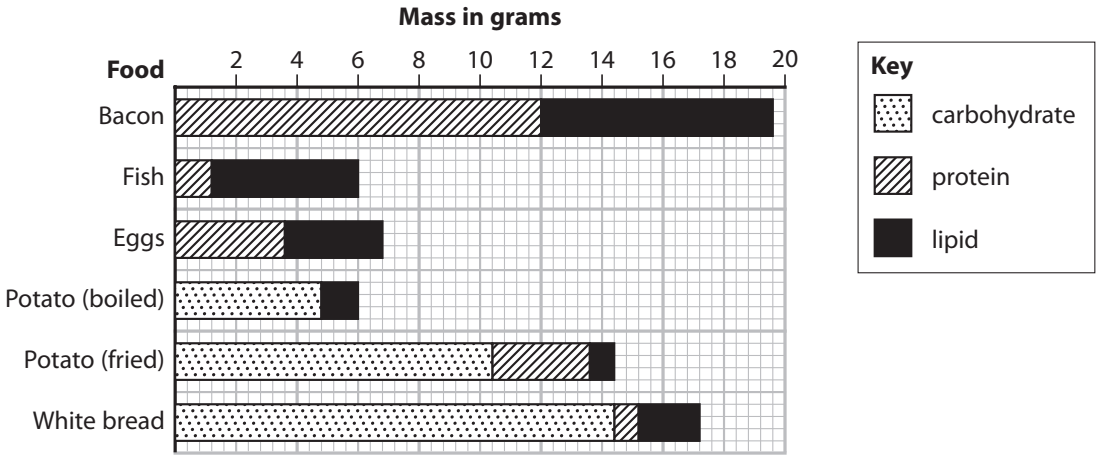
# 4. Question 7 · part a.i · 1 mark

January 2022 Paper 1

Further practice · Attempt part a.i only.

**7** Most diets contain carbohydrates, lipids and proteins.

The bar chart shows the masses of each of these in 26 g samples of different foods.




(a) (i) State the names of the three chemical elements found in all carbohydrates, lipids and proteins.

**(1)**

.....

(ii) State the name of one chemical element found in proteins but not found in carbohydrates or lipids.

**(1)**

.....

(iii) Deduce the mass of lipid in 26 g of bacon.

**(2)**

mass of lipid = ..... g

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## 4. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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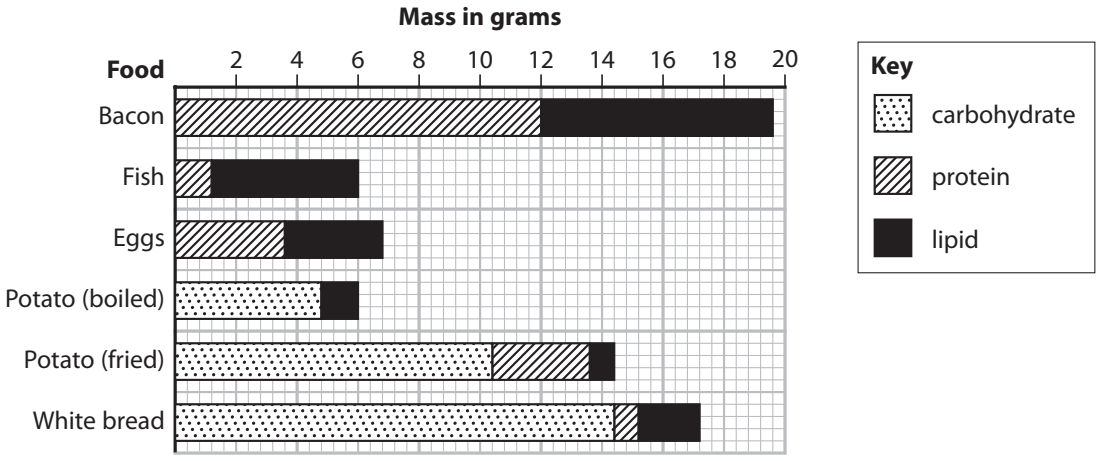
# 5. Question 7 · part a.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part a.ii only.

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P 6 7 0 5 9 A 0 2 0 2 4

## 5. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

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# 6. Question 5 · part a · 2 marks

June 2022 Paper 2R

Further practice · Attempt part a only.

**5** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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## 6. Question 5 · part a · 2 marks · Working and answer

Working and answer

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## 7. Question 4 · part a.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

**4** Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

**(2)**

.....

.....

.....

.....

(ii) Name the hormone that stimulates the liver to produce glycogen.

**(1)**

.....

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

**(1)**

.....

(b) Consuming a diet high in carbohydrates can increase the BMI of a person.

The table gives information on the BMI categories of individuals.

| BMI | Category |
|---|---|
| less than 18.5 | underweight |
| 18.5 to 24.9 | healthy weight |
| 25 to 29.9 | overweight |
| 30 to 34.9 | obese |
| 35 to 39.9 | severely obese |
| more than 39.9 | morbidly obese |

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## 7. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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